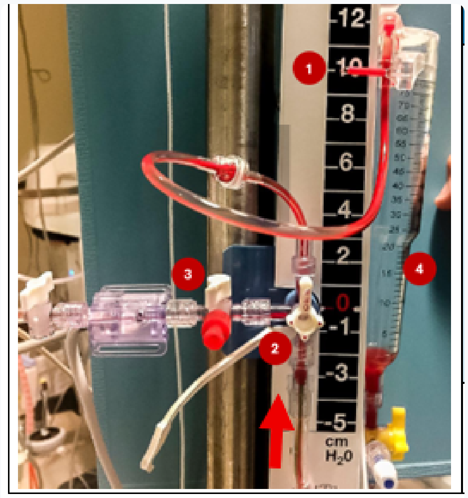
External Ventricular Drain



COMPONENTS

1. **Drainage setting:** Increments of 5 cmH_2O, higher = less drainage (higher resistance/back-pressure).
2. **Drainage stopcock:** 12 o'clock = clamp/closed, 3 o'clock = open to drain.
3. **Transducer and zeroing stopcock:** Controls baseline calibration.
4. **Collection/drip chamber:** Graduated cylinder measuring CSF volume.

**Red arrow indicates the direction of CSF flow.*

How to read the number: the cmH_2O setting is the drainage threshold above the tragus/EAM. Lower or less-positive settings drain more easily; higher or more-positive settings drain less unless ICP exceeds that higher threshold.

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INDICATIONS

- **CSF Diversion** for acute obstructive hydrocephalus (e.g., IVH, posterior fossa stroke).
- **ICP Monitoring** in severe brain injury (GCS ≤ 8).

SIGNS OF HYDROCEPHALUS

Clinical Signs:

- Decline in Level of Consciousness (LOC) or progressive somnolence.
- **Parinaud's Syndrome:** Upward gaze palsy (setting sun sign), retraction nystagmus on convergence, and pupillary light-near dissociation.

Radiographic Signs (NCCT Head):

- Progressive enlargement of the cerebral ventricles.
- Temporal horn dilation (sensitive early sign of obstruction).
- High-risk factors: Intraventricular Hemorrhage (IVH) in 3rd or 4th ventricles, compression of 4th ventricle, or high volume blood.

BASICS

- **Leveling:** Always align the zero level of the EVD scale/transducer to the external auditory meatus (EAM) / tragus.
- **Mobilization Clamping Rules:** Always CLAMP the EVD before: turning the patient, adjusting HOB, or mobilizing the patient out of bed to prevent severe overdrainage or underdrainage.
- **Waveform:** ICP value and waveform morphology are valid only when the EVD is clamped.
- **CSF Drainage:** CSF drainage is passive: occurs only when patient ICP exceeds EVD chamber height. Example: an EVD at 5 cmH_2O drains more readily than one at 20 cmH_2O.
- **Normal CSF Flow:** Normal CSF production is ~20 mL/hr (~500 mL/day). Drainage >20 mL/hr should trigger immediate assessment for overdrainage or chamber level escalation.
- **Weaning:** Gradual escalation of drainage setting by 5 cmH_2O per day. After +20 cmH_2O, EVD should be clamped & head CT obtained to evaluate ventricular caliber. Neurologic examination, CSF output, and ICP waveform should be assessed daily.

COMPLICATIONS

- **Overdrainage (>20 mL/hr):** Risk of subdural hematomas (bridging vein tearing), ventricular collapse (slit ventricles), or upward cerebellar herniation.
- **Underdrainage:** Risk of worsening hydrocephalus, brain compression, or elevated ICP. Troubleshoot for system kinks, blood clots, air locks, or malpositioned stopcocks.